Sage Clokey

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OBJECTIVE

Writer, content creator, and bioinformatics researcher making the case for liberty through the lens of biology. Author of a 126-chapter trilogy connecting Austrian economics, molecular biology, and the philosophy of conscience. Seeking to develop editorial and multimedia skills to communicate what the biological record proves: that decentralized, voluntary order is not merely a political ideal — it is how life itself works.

EDUCATION

University of California, Santa Cruz B.S. in Biomolecular Engineering and Bioinformatics — Bioinformatics Concentration, 2026

Relevant Coursework: Data Science, Machine Learning, Statistical Modeling, Genomics, Systems Biology, Computational Biology, Bioethics, Applied Econometrics

WRITING & CONTENT CREATION

The Living Age Trilogy — Author | 2025–Present - Three-volume work (126 chapters) mapping Austrian economics onto molecular biology, media mythology, and the architecture of conscience - Covers Mises' calculation problem in cellular metabolism, Hayek's knowledge problem in gene regulatory networks, and Menger's theory of money in ATP emergence - Includes extended media analysis (Star Wars, Lord of the Rings, Avatar, The Matrix) as modern mythology encoding the spiral-vs-block conflict

"The Living Architecture" — Capstone Research Paper | Spring 2026 - Three-layer computational analysis proving biological systems operate as free economies, not command hierarchies - Submitted for publication to Frontiers in Systems Biology - Key findings: 19:1 robustness ratio favoring decentralized architecture; distributed metabolic allocation retains 71% output vs. 53% centralized; FBA models miss 30% of knowledge that is structurally local

"The Bones of Joseph" — Essay | 2026 - Competition essay connecting Clokey family legacy, Austrian economics, and biblical theology - Awarded and published

Published Essays & Drafts - "The Distributed Intelligence of Life" — companion essay on biological decentralization - "The Living Republic of Conscience" — declaration of voluntary republicanism - "Isolation Is the Disease" — essay on social genomics and the knowledge problem - "The Economy of Stewardship" — anti-usury economics through a biological lens

YouTube Channel Development — The Living Age | In Development - Four-week content series: The True Republic, The Sagent Creed, Media Mythology, The Eight Pillars - Bridge episodes connecting bioinformatics research to philosophy of liberty

LIBERTY MOVEMENT EXPERIENCE

Students for Liberty — California State Director | 2025–Present - Lead statewide campus outreach connecting STEM students to the philosophy of liberty through biology - Developed Living

Age Impact Project proposal — cross-disciplinary program using biological science and creative media - Organize Socratic discussion events, manage outreach pipeline across California campuses

FEE Summer Campus | July 2026 - Presented thesis connecting synthetic biology and libertarianism

SynBioBeta — Attendee & Networker | June 2026 - Largest synthetic biology conference in North America - Connected with leaders in synbio, policy, and publishing (Drew Endy, Andrew Hessel, Senate Biotech Caucus)

Cato University — Cato Institute | Summer 2025 & February 2026 - Attended twice by competitive selection; intensive seminars on political philosophy and economics

Young Americans for Liberty — State Chair, California | 2025 - Managed statewide network of student chapters; coordinated data-driven outreach across multiple campuses

Leadership Institute — Gulf of America Cruise | Fall 2025 - Selective leadership training for emerging liberty movement leaders

RESEARCH & TECHNICAL PROJECTS

Living Systems as Decentralized Economies — BME 129C Capstone | Spring 2026 - Analyzed E. coli gene regulatory network (~4,500 interactions) proving robustness of decentralized architecture - Modeled KEGG metabolic pathways as economic agents; computed Nash equilibrium under distributed vs. centralized allocation - Mapped codon usage across organisms as trade friction, revealing comparative advantage patterns

The Living Republic — Political Topology via PCA - Applied PCA to voting records of 1,091 Congressional members and 199 UN nations - Demonstrated political space is continuous and multidimensional; published as interactive data visualization

Biology & Voluntaryism — Evolutionary Cooperation Analysis - Interactive D3.js visualization analyzing 4-billion-year cooperation patterns in living systems

LivingWorks — Genome Design Platform | In Development - Life-based design system treating biological engineering as stewardship of living order

TECHNICAL SKILLS

Languages: Python, R, JavaScript/React, Java, C, SQL, Bash/Shell

Data Science & ML: PCA, UMAP, clustering, OLS regression, LASSO, Random Forest, data visualization (D3.js, Chart.js, ggplot2)

Bioinformatics: Seurat, Scanpy, BLAST, UCSC Genome Browser, Bioconductor, NetworkX; single-cell RNA-seq, population genetics

Content Tools: Markdown, WeasyPrint, HTML/CSS publishing, GitHub Pages

ADDITIONAL INFORMATION

Portfolio: sage-clokey.github.io **Lineage:** Grandson of Art Clokey, creator of Gumby (Premavision Studios) **Interest Areas:** Science communication, editorial writing, video production, economics and biology, decentralized systems